INSTITUTO COLLEGE

MATH 110: General Education Math

Fall 2026

Exam 1 Review Worksheet

Name: _____

This worksheet is ungraded. It has more problems than we can get to in class, so use the rest for practice at home.

Counting

1. Let $U = \{1, 2, 3, \ldots, 12\}$ be the universal set. Let A be the set of multiples of 3 in U, and let B be the set of odd numbers in U.

   (a) Write A and B by listing their elements.

   (b) Find $A \cup B$, $A \cap B$, and A^c.

   (c) Find $(A \cup B)^c$ and $|A \cup B|$.

   (d) Are A and B disjoint? Explain.

2. A sandwich shop lets you choose one of 4 breads, one of 5 fillings, and one of 3 cheeses, or no cheese at all. How many different sandwiches are possible?

3. Compute each of the following without a calculator.

(a) $\dfrac{9!}{7!\,2!}$

(b) $0! + 1!$

(c) $(2+3)!$ and $2! + 3!$. Are they equal?

4. Eight runners compete in a race. Gold, silver, and bronze medals go to the first, second, and third place finishers. In how many ways can the medals be awarded, assuming there are no ties?

5. Consider the different arrangements of the five letters of the word NURSE, such as RUNES and SNUER.

(a) How many arrangements are there?

(b) How many arrangements begin with a vowel?

(c) How many arrangements begin with N and end with E?

6. A pizza place offers 10 different toppings. How many different pizzas with exactly 3 different toppings are possible?

7. You are making a playlist of 4 different songs, chosen from your 9 favorites, and the order of the songs matters.

(a) How many playlists are possible?

(b) Find the answer to part (a) a second way: first choose which 4 songs to use, and then put them in order.

8. A nurse educator will show 3 different training videos to new staff, one after the other, chosen from 6 videos on infection control and 5 videos on patient safety. The first video shown must be on infection control.

(a) How many different lineups of 3 videos are possible?

(b) How many of these lineups include at least one video on patient safety?

9. Four friends each tell you the month they were born. How many lists of four birth months, one for each friend, have at least two friends born in the same month?

10. Of the 50 volunteers at a hospital, 28 are certified in CPR, 19 are certified in first aid, and 12 are certified in neither.

(a) How many of the volunteers are certified in both?

(b) How many are certified in exactly one of the two?

11. How many arrangements of the letters A, B, C, D, and E have A first or B last?

12. A patient-safety committee will have a chair and 4 other members with no distinct roles, all chosen from 8 nurses and 4 pharmacists. No one can fill more than one position, and the chair must be a pharmacist.

(a) How many committees are possible?

(b) How many committees have only nurses among the 4 other members?

(c) How many committees have at least one pharmacist among the 4 other members?

13. How many 4-digit whole numbers (from $1{,}000$ to $9{,}999$) have no repeated digit and are even?

14. Six patients sit in a row of 6 chairs in a clinic waiting room. Two of them, Mr. Lopez and Ms. Grant, do not want to sit next to each other. In how many ways can the six patients be seated?

Probability

15. A spinner is divided into three equal sections colored red (R), green (G), and blue (B). The spinner is spun twice, and the colors are recorded in order.

(a) Write the sample space by listing its sample points.

(b) Write the event A, "both spins land on the same color," and the event B, "at least one spin lands on red," by listing their sample points.

(c) The sample points are equally likely. Find $P(A)$ and $P(B)$.

16. A die is weighted so that it lands on 6 with probability $\frac{1}{2}$, and on each of the numbers 1 through 5 with the same probability c.

(a) Use the fact that $\sum_{x \in S} P(x) = 1$ to find c.

(b) Let $E = \{1, 3, 5\}$ be the event "the die lands on an odd number." Write $P(E)$ using sigma notation, then evaluate it.

(c) Let $F = \{5, 6\}$ be the event "the die lands on a number greater than 4." Write $P(F)$ using sigma notation, then evaluate it.

17. Two fair dice are rolled. What is the probability that the larger of the two numbers the dice land on is 3? (If both dice land on the same number, that number counts as the larger one.)

18. A hospital has 10 volunteers, 6 of whom are volunteering for the first time and 4 of whom are returning. Three of the volunteers are chosen at random to appear in a newsletter photo.

(a) What is the probability that all 3 are returning volunteers?

(b) What is the probability that exactly 2 are returning volunteers?

(c) What is the probability that at least one is a returning volunteer?

19. A spinner lands on red with probability $\frac{1}{2}$, on blue with probability $\frac{1}{3}$, and on green with probability $\frac{1}{6}$. It is spun twice, and the spins are independent.

(a) What is the probability that both spins land on the same color?

(b) What is the probability that the two spins land on different colors?

(c) What is the probability that one spin lands on red and the other on blue?

20. Let A and B be events with $P(A) = 0.6$ and $P(B) = 0.5$. The probability that neither A nor B occurs is 0.2.

(a) Find $P(A \cup B)$.

(b) Find $P(A \cap B)$.

(c) Find $P(A \cap B^c)$, the probability that A occurs but B does not.

(d) Are A and B mutually exclusive? Are they independent? Explain.

21. A commuter takes a bus, then a train, then a shuttle to get to work. The bus is on time with probability 0.8, the train with probability 0.9, and the shuttle with probability 0.95, independently of each other.

(a) What is the probability that all three are on time?

(b) What is the probability that at least one of them is late?

(c) What is the probability that the bus is late, but the train and the shuttle are on time?

22. A hospital lab mislabels each blood sample with probability 0.002, independently of the other samples. On a busy day, the lab processes 500 samples. What is the probability that at least one of them is mislabeled? (You may use a calculator.)

23. A drawer holds 6 black socks and 4 white socks. You pull out 2 socks at random, one after the other, without looking. What is the probability that the two socks match?

24. A store records how 200 of its customers pay and whether they use a coupon.

	Uses a coupon	No coupon	Total
Pays by card	48	92	140
Pays with cash	12	48	60
Total	60	140	200

One of the 200 customers is chosen at random. Let K be the event "the customer pays by card" and Q the event "the customer uses a coupon."

(a) Find $P(Q)$, $P(Q \mid K)$, and $P(Q \mid K^{\mathrm{c}})$.

(b) Find $P(K \mid Q)$, and describe it in words.

(c) Are K and Q independent? Explain.

25. A blood bank gets 60% of its units from Donor Center A and 40% from Donor Center B. Screening rejects 2% of the units from Center A and 5% of the units from Center B.

(a) Make a two-way table for a group of $1{,}000$ units, with the donor center for the rows and whether the unit is rejected for the columns.

(b) What is the probability that a randomly chosen unit is rejected?

(c) A unit is rejected. What is the probability that it came from Center B?

26. A fair die is rolled. Let $A = \{1, 2\}$, $B = \{2, 4, 6\}$, and $C = \{5, 6\}$. For each pair of events, decide whether the events are mutually exclusive and whether they are independent.

(a) A and B

(b) A and C

(c) B and C

27. Two fair dice are rolled. Let A be the event "the first die lands on an odd number" and B the event "the numbers the dice land on add up to at least 9." Find $P(B \mid A)$, and compare it with $P(B)$. Are A and B independent?

28. A basketball player shoots two free throws. She makes the first one with probability 0.7. If she makes the first, she makes the second with probability 0.8; if she misses the first, she makes the second with probability 0.5.

(a) What is the probability that she makes both?

(b) What is the probability that she makes exactly one?

(c) What is the probability that she misses both?

29. Each seed in a packet sprouts with probability 0.85, independently of the others. You plant 6 of the seeds.

(a) What is the probability that all 6 sprout?

(b) What is the probability that exactly 5 sprout?

(c) What is the probability that at most 4 sprout?

30. A clinic's front desk calls patients the day before their appointments. Each patient answers the phone with probability 0.6, independently of the others. The desk calls 5 patients.

(a) Explain why this is a binomial experiment. What are n, a success, and p?

(b) What is the probability that exactly 3 of the patients answer?

(c) What is the probability that at least one of the patients answers?

31. A couple plans to have 4 children. Assume that each child is equally likely to be a boy or a girl, independently of the others.

(a) What is the probability that they have exactly 2 girls?

(b) What is the probability that they have at least one boy and at least one girl?